2.3 Solving Trigonometric Equations (day 2)

Name _____ Period ____ Date _____

Pre-Calculus 2.3 (30-minute plan) · Student sheet

I can…

- use inverse trig functions to solve a trig equation and find all of its solutions.

Vocabulary · fill in the missing word

Inverse trig function — _____, arccos, arctan: give the angle for a given value ($\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$ on the calculator)

Range of arcsin / arccos / arctan — $[-\pi/2, \pi/2]$, $[0, \pi]$, $(-\pi/2, \pi/2)$: the _____ only gives answers there

Reference angle — the acute angle to the _____; use it to find the solution in the other quadrant

Approximate solution — a _____ answer (4 decimal places in radians) when the value isn't a unit-circle value

Part 1 · Notes and examples (fill in each note, then work the example)

Where the calculator answer lands: arcsin gives an angle in _____; arccos gives an angle in _____; arctan gives an angle in _____.

The second solution: sin x = k: x and _____ cos x = k: x and _____ tan x = k: x and _____

Negative answers: A negative calculator answer? Add _____ (or _____ for tangent) to land in $[0, 2\pi)$. Calculator in _____ mode!

Example 1

Solve on $[0, 2\pi)$: $\sin x = 0.3$

Example 2

Solve on $[0, 2\pi)$: $\cos x = -0.4$

Example 3

Solve on $[0, 2\pi)$: $3\sin^2 x - 5\sin x - 2 = 0$

Part 2 · Partner practice

Partner practice

Solve on $[0, 2\pi)$: $2 \cos x = 1.5$

Part 3 · On your own: choose 2 of 3

Problem 1

Solve on $[0, 2\pi)$ (round to 4 decimals): $\sin x = 0.7$

Problem 2

Solve on $[0, 2\pi)$ (round to 4 decimals): $\tan x = -2$

Problem 3

Solve on $[0, 2\pi)$ (round to 4 decimals): $2 \sin^2 x + 5 \sin x - 3 = 0$